

Vinayak Kamankar

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Career Objective / Summary

B.Tech student specializing in Artificial Intelligence and Machine Learning, with a strong background in IoT, Embedded Systems, and Web Development. Proficient in Python, Java, and Web Technologies with hands-on project experience in smart systems. Passionate about building intelligent and sustainable tech solutions. A quick learner with a problem-solving mindset and strong teamwork and leadership abilities.

Academic Credentials

Bachelor of Technology – AIML Sanjivani University, Kopergaon	2024 – 2027
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Diploma in Mechatronics – 80% Government Polytechnic, Nashik	2021 – 2024
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Skills

Technical Skills:

- Programming Languages: Python, Java, HTML, Java Script , CSS
- Embedded Systems Development
- Microcontroller Programming
- CAD & Engineering Design

Soft Skills:

- Leadership & Teamwork
 - Problem Solving
 - Creative Thinking
 - Communication
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Projects

1. Smart Traffic Management System using IoT & Computer Vision

- **Objective:** To reduce urban traffic congestion and enhance road safety through real-time adaptive signal control and automated violation detection.
 - **Technologies:** NodeMCU, Ultrasonic Sensors, IR Sensors, Python, OpenCV, ANPR (Automatic Number Plate Recognition) Camera, Cloud Services.
 - **Features:**
 - Implemented dynamic signal timing that adapts to real-time vehicle density.
 - Developed a computer vision module for automated red-light violation detection and license plate logging.
 - Designed a cloud-based dashboard for monitoring traffic flow and violation data.
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2. AgroSense – Smart Crop Analysis & AI Agronomy Assistant

- **Objective:** To provide farmers with data-driven agricultural decisions by recommending optimal crops using ML and offering expert consultation on pest control and farm management using a multimodal AI.
 - **Technologies:** Python, Flask, Scikit-learn (Random Forest), Pandas, Google Gemini API, HTML, CSS, JavaScript.
 - **Features:**
 - Developed an ML model (RandomForestClassifier) that recommends one of 22 crops by analyzing 7 soil/environmental factors (N, P, K, pH, temperature, humidity, rainfall).
 - Integrated a Gemini-powered AI assistant for expert advice on irrigation, fertilizers, and pest management.
 - Implemented a multimodal "AI Image Recognition" feature, allowing users to upload a photo of a diseased plant for an AI-based diagnosis and treatment recommendation.
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3. Fire Fighting Robot with Web Integration

- **Objective:** To design and build an autonomous robot capable of detecting and suppressing fires in indoor environments, with a web-based manual override.
 - **Technologies:** ESP32-CAM, Flame Sensor, Ultrasonic Sensors, Motor Driver, Web Interface (HTML/JS), WebSocket.
 - **Features:**
 - Programmed autonomous navigation with obstacle avoidance using ultrasonic sensors.
 - Enabled real-time fire detection using flame sensors, triggering an automated water pump.
 - Created a web-based UI for live video streaming and manual remote control of the robot's movement and water cannon.
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Internship

Maceltron Machines Pvt. Ltd., Nashik

Electrical Intern | Jan 2024 – Apr 2024

- Assisted in designing and wiring control panels.
 - Tested and debugged PLC-based automation systems.
 - Handled documentation and setup for industrial machines.
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Certifications

- Power BI Data Modelling Basics
 - SQL and Relational Database 101 – IBM
 - Building Cloud and Multicloud Applications – IBM
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Achievements

IIT Guwahati Event (Techniche) - First Rank in Micromouse Competition (Maze Solving Robot)

Extra-Curricular Activities

- Vice President of the Design & Thinking Club under the Innovation Cell .
 - Cricket: Active member of the college cricket team. Participated in multiple inter-departmental and college-level tournaments.
 - Volunteered in tech fests and workshops.
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Personal Details

- Date of Birth: November 4, 2005
 - Address: Jakhori, Nashik, Maharashtra
 - Languages Known: English, Hindi, Marathi
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Strengths

- Quick Learner and Adaptable
- Strong Work Ethic
- Innovative Thinking
- Team Player and Collaborative